MVA Hackathon 2026 – Track 2: Drug Repurposing Report

Restoring BUBR1 dosage and buffering aneuploidy stress in a BUB1B compound-heterozygous MVA1 case

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> **Disclaimer.** All candidates in this report are hypotheses for follow-up experiments. They are not evidence that any medicine works for this child, and no dosing is proposed.

1. Summary

The proband carries two rare variants in **BUB1B** (Track 1): **p.Leu737Ter**, a premature stop codon expected to cause nonsense-mediated decay (null allele), and **p.Asn1002Lys**, a missense variant absent from gnomAD inside the folded C-terminal kinase-like domain (hypomorphic allele). The combined effect is a **partial loss of function: less BUBR1 protein, and the remaining protein is probably less stable**. This weakens the spindle assembly checkpoint (SAC) and causes mosaic aneuploidy, growth restriction and cancer predisposition (rhabdomyosarcoma in this child).

Open Targets shows that BUB1B has strong genetic links to MVA but **no approved or clinical-stage drug**, and the ligands that exist would inhibit the protein, the opposite of what this patient needs. We therefore used **indirect strategies** and grouped seven candidates into three tiers:

Tier	#	Candidate (approved drug unless noted)	Strategy
1. Restore BUBR1	1	**NAD+ precursors** (nicotinamide, niacin; nicotinamide riboside as supplement)	Slow BUBR1 degradation via SIRT2
	2	**Chemical chaperone** (sodium phenylbutyrate; ursodiol)	Help the N1002K protein fold
	7	**Rational combination** of 1 + 2	Protect the only protein-producing allele from two sides
2. Buffer consequences	3	**Senolytic** (dasatinib, ± quercetin)	Clear senescent cells
	5	**Low-dose sirolimus**	Support autophagy / mitophagy
	6	**N-acetylcysteine**	Reduce oxidative stress
3. Exploratory	4	**Metformin / hydroxychloroquine**	Selectively stress aneuploid cells

The lead hypothesis is **candidate 1**, with **candidate 7** as the most patient-specific idea. We also flag drugs to avoid or monitor (Section 7) and propose concrete experiments (Section 8).

2. Case and phenotype

The clinical document lists eight HPO terms: short stature (HP:0004322), failure to thrive (HP:0001508), skeletal muscle atrophy (HP:0003202), premature birth (HP:0001622), small for gestational age (HP:0001518), parental recurrent spontaneous abortion (HP:0200067), rhabdomyosarcoma (HP:0002859) and nephrocalcinosis (HP:0000121). Read together, as the organisers recommend, they describe a chromosomal-instability disorder: cancer predisposition, pre- and postnatal growth restriction, renal involvement and reproductive loss in the family. This is consistent with MVA type 1, in which biallelic BUB1B variants and childhood cancers, including embryonal rhabdomyosarcoma, are reported (Hanks et al., 2004).

Two phenotype features were used to guide drug selection:

- **Cancer predisposition with a history of rhabdomyosarcoma.** Any candidate that may promote growth of existing tumours is flagged, and growth-promoting therapies are listed under "avoid / use with caution".
- **Muscle atrophy and failure to thrive.** These overlap with the progeroid phenotype of BubR1-deficient mice (sarcopenia, fat loss), which motivates the tier 2 candidates.

3. Variant mechanism

3.1 The two alleles

	Variant 1	Variant 2
HGVS (ENST00000287598)	p.Leu737Ter	p.Asn1002Lys
Consequence	stop_gained, exon 17/23	missense, exon 23/23
gnomAD	max AF 9.98×10^{-5}	absent
Predictions	expected NMD (stop not in last exon)	SIFT deleterious (0.01), PolyPhen probably damaging (0.997)
Predicted effect	**null allele**	**hypomorphic allele** (protein made, reduced function / stability)

The two variants have not been phased (no parental data), so compound heterozygosity is assumed, not proven. A null allele combined with a hypomorphic missense allele is the typical MVA1 configuration, because complete loss of BUBR1 is generally not compatible with life.

3.2 Where N1002K sits in the protein

BUBR1 (UniProt O60566, 1,050 residues) has an N-terminal TPR domain, a long disordered middle region with short binding motifs, and a C-terminal kinase-like domain. In the AlphaFold model (AF-O60566-F1, v6) the average pLDDT is only 63.6, because about 37% of residues are predicted to be disordered. The predicted aligned error (PAE) map, however, shows that **residues ≈720–1050 form one compact, confidently predicted domain**: the mean PAE inside this block is 8.1 Å, compared with 26.6 Å elsewhere, and its mean pLDDT is 82.0. **N1002 lies inside this folded domain, with a pLDDT of 91.1 (very high confidence)**, whereas p.Leu737Ter truncates the protein just before it (Figure 3; values from `track2/evidence_check.py`).

A charge-changing substitution (neutral asparagine → positively charged lysine) inside a compact folded domain, predicted damaging by SIFT and PolyPhen, is a common way for missense variants to reduce protein stability. BUBR1 levels are also known to be controlled by ubiquitin-mediated degradation (Section 4; North et al., 2014). Our **working hypothesis** is therefore that **N1002K destabilises the kinase-like domain, so the protein is degraded faster and BUBR1 levels drop further** than expected from one remaining allele. This is an inference, not a measurement, and it is directly testable (Section 8).

3.3 Mechanism characterisation

Required element	Finding
Loss- or gain-of-function	**Partial loss of function** (reduced BUBR1 dosage): one null allele + one hypomorphic allele. No evidence of gain of function.
Pathway disrupted	**Spindle assembly checkpoint (SAC)**. BUBR1 is part of the mitotic checkpoint complex (with BUB3, MAD2 and CDC20) that inhibits the APC/C until all chromosomes are correctly attached.
Downstream biological consequence	Chromosome mis-segregation → **mosaic aneuploidy** → aneuploidy stress and senescence → **growth restriction, muscle atrophy** and **cancer predisposition** (details below).

Downstream chain, step by step: 1. Premature anaphase and chromosome mis-segregation → **mosaic aneuploidy**. 2. Aneuploid cells carry extra protein load and develop **proteotoxic stress, mitochondrial dysfunction and oxidative stress**, which drive **senescence** (Joy et al., 2021). 3. Loss or senescence of aneuploid cells in growing tissues → **growth restriction and muscle atrophy**. BubR1-deficient mice show progeroid features, including sarcopenia, that improve when senescent cells are removed (Baker et al., 2011). 4. Survival of aneuploid cells with a weak checkpoint → **cancer predisposition** (rhabdomyosarcoma in this child).

Each of these steps is a point where a drug could act, and this structure defines our candidate tiers.

4. Is there a direct drug? (Open Targets)

We queried the Open Targets Platform for BUB1B (ENSG00000156970; accessed 24 October 2026, web interface).

- **Associations (Figure 1).** BUB1B–MVA has strong evidence from genetic sources (ClinVar, GEL PanelApp, Gene2Phenotype, UniProt, Orphanet, ClinGen), but the **Clinical Precedence** column (drugs in trials or approved for this target–disease pair) is **empty**. The API confirms this for both MVA entries (MONDO_0000141, association score 0.81; MONDO_0009759, MVA type 1, score 0.79): their evidence comes only from genetic association and literature, with no drug or clinical evidence.
- **Tractability (Figure 2).** No approved or clinical-stage drug in any modality (small molecule, antibody, PROTAC, other). BUB1B is flagged as a *druggable family* with *high-quality ligands*, but such ligands would bind or inhibit BUBR1. Since this patient has **too little** BUBR1, inhibition is the wrong direction.
- **Ubiquitination evidence.** The PROTAC section flags ubiquitination of BUBR1 in UniProt and other databases. This independently supports the idea that BUBR1 levels are controlled by ubiquitin-mediated degradation, which candidates 1 and 2 aim to reduce.

Conclusion: there is no drug that directly restores BUBR1 function, so all candidates act indirectly, either on BUBR1 stability or on the downstream consequences.

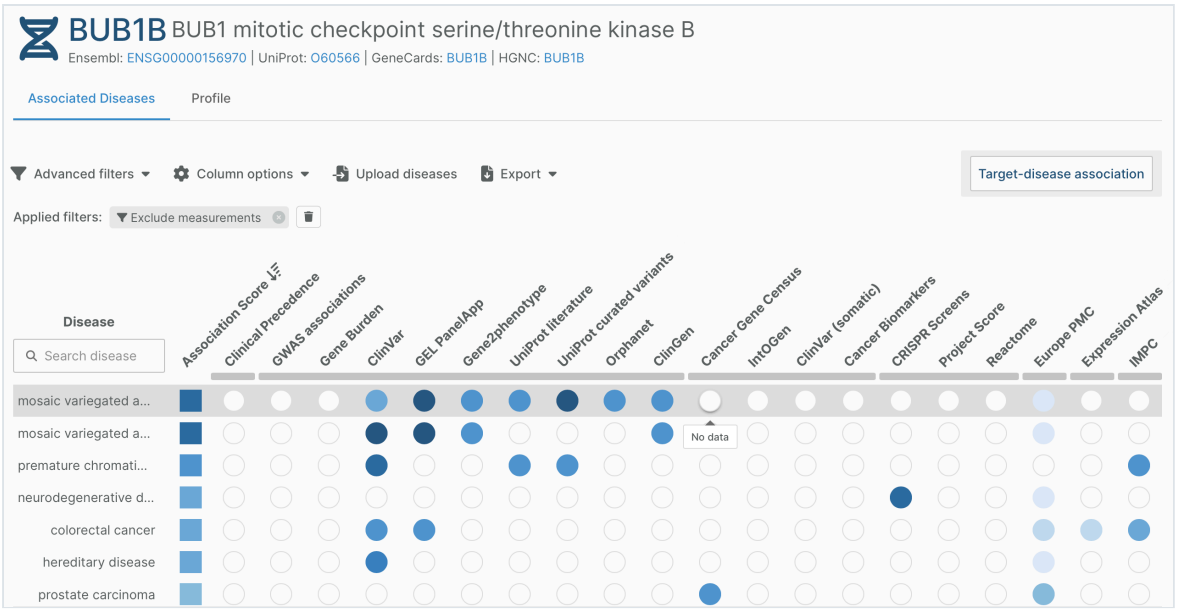


Figure 1. Open Targets: BUB1B disease associations. MVA rows show strong genetic evidence and no Clinical Precedence (no drugs).

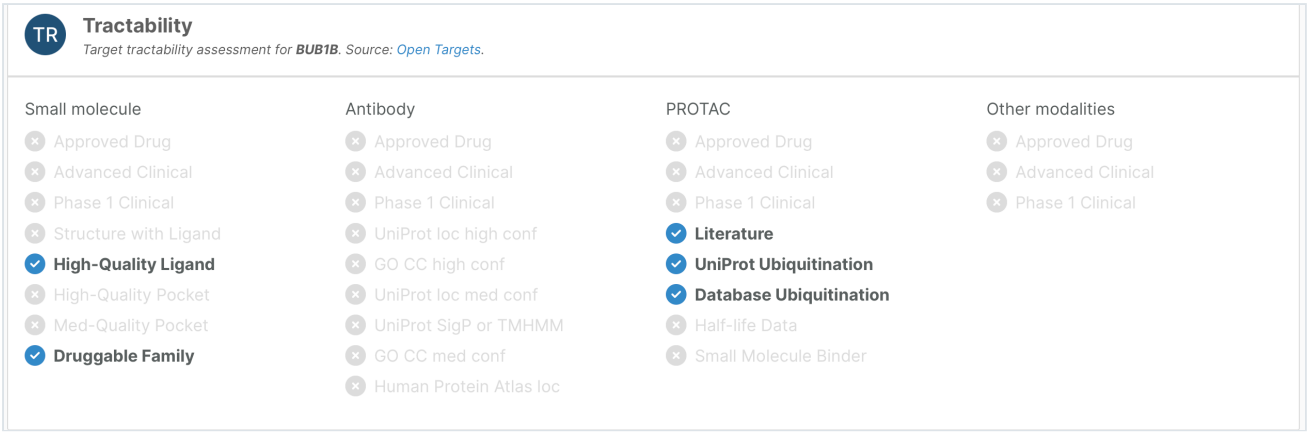


Figure 2. Open Targets: BUB1B tractability. No approved or clinical drugs in any modality; ubiquitination evidence under PROTAC.

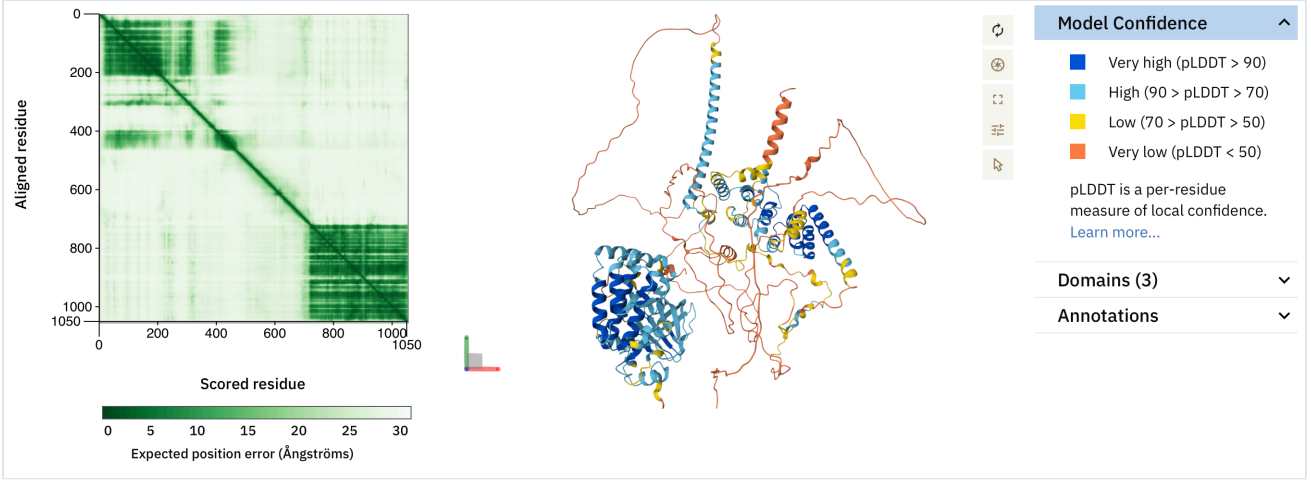


Figure 3. AlphaFold model AF-O60566-F1. Left: PAE map; the dark block at ≈720–1050 is a compact, confidently predicted C-terminal domain containing N1002. Right: model coloured by pLDDT.

5. Methods

1. **Mechanism characterisation.** Track 1 annotation (Ensembl VEP 116 with gnomAD, SIFT and PolyPhen), AlphaFold structure and PAE, UniProt, Open Targets and literature on BUBR1 structure and MVA.
2. **Druggable-point mapping.** We listed each step of the mechanism chain (Section 3.3) and asked, for each, whether an approved drug can modify it.
3. **Candidate search.** Open Targets (target, tractability, known drugs), targeted PubMed / literature review on BUBR1 regulation, aneuploidy stress biology and senescence, and drug labels and clinical-trial records for safety and paediatric use. We attempted to use TxGNN (knowledge-graph repurposing model) as an independent check, but its web interface was not reachable during the analysis; this is noted as a limitation.
4. **Evidence grading.** Each candidate was scored on the same five criteria (Section 6, Table 2), and drugs with a plausible link but unacceptable risk were recorded as rejected (Section 7) rather than silently dropped.

This was a manual, literature-driven curation, not an automated screen. The grading scheme in Table 2 is designed to be reusable for other variants (Section 10).

Reproducibility. The database evidence in Section 4 is reproduced by a script in the GitHub repository (`track2/evidence_check.py`, Python standard library only). It queries the Open Targets GraphQL API for BUB1B disease associations, evidence types and tractability, and downloads the AlphaFold model and PAE file to report the confidence at residue 1002 and the PAE within the C-terminal block. The candidate list and the grading in Table 2 are stored as a table (`track2/candidates.csv`), so every candidate, its evidence level and its flags can be inspected or re-scored.

6. Candidates

Candidate 1 (lead): NAD+ precursors — slow BUBR1 degradation

- **Rationale.** BUBR1 is targeted for degradation by the APC/C–CDC20 ubiquitin ligase. North et al. (2014) reported that the NAD+-dependent deacetylase **SIRT2** deacetylates BUBR1 and protects it from this degradation; raising NAD+ with the precursor NMN increased BUBR1 levels in aged mice, and SIRT2 overexpression extended lifespan in BubR1 hypomorphic mice. This patient has a hypomorphic allele, so *some* BUBR1 is made; slowing its degradation could raise its level.
- **Drugs.** Approved: **nicotinamide** and **niacin** (vitamin B3). Best clinical data: **nicotinamide riboside (NR)**, a dietary supplement.
- **Safety precedent.** NR has been given for 18 months to patients with **ataxia telangiectasia** (most of them children), another genome-instability syndrome with cancer predisposition, without notable adverse events.
- **Risks.** In a small mouse study, NR increased metastasis of an existing breast-cancer xenograft (Maric et al., 2022). Given the child's rhabdomyosarcoma, NAD+ supplementation should only be considered in remission and under oncology supervision. SIRT2's role in cancer is context dependent. Niacin causes flushing and liver toxicity at high doses.
- **Additional link.** NAD+ also supports mitophagy, so this candidate may act on tier 2 biology as well.

Candidate 2: chemical chaperone — help N1002K fold

- **Rationale.** If N1002K destabilises the kinase-like domain (Section 3.2), a chemical chaperone could improve folding and reduce degradation of the only BUBR1 protein this patient makes. This rationale is **specific to this variant type** (a missense variant in a folded domain) and would not apply to a patient with two null alleles.

- **Drugs. Sodium phenylbutyrate (4-PBA)**, approved for urea-cycle disorders including in children, and a known chemical chaperone. **Ursodiol (UDCA)**, approved bile acid with chaperone-like activity (TUDCA is its taurine conjugate).
- **Risks / open questions.** 4-PBA is also a histone deacetylase inhibitor, so it changes protein acetylation. Because candidate 1 depends on BUBR1 *deacetylation*, the interaction between the two must be tested.

Candidate 7: rational combination of 1 + 2

The patient has a single protein-producing allele. Candidates 1 and 2 protect that protein from two sides: correct folding (chaperone) and reduced ubiquitin tagging (NAD+/SIRT2). The rationale comes directly from the genotype, which makes this the most patient-specific hypothesis in the report. The 4-PBA acetylation caveat above applies.

Candidate 3: senolytic — clear senescent cells

- **Rationale.** In BubR1 progeroid mice, genetic clearance of p16Ink4a-positive senescent cells delayed the onset of sarcopenia, fat loss and cataracts (Baker et al., 2011). This is *in vivo* and BubR1-specific evidence that matches this child's muscle atrophy and failure to thrive.
- **Drug. Dasatinib** (approved, including paediatric indications), studied as a senolytic with quercetin.
- **Risks.** Dasatinib is a cytotoxic kinase inhibitor with significant side effects. Baker et al. used genetic, not pharmacological, clearance. This is a long-term hypothesis.

Candidate 5: low-dose sirolimus — support autophagy / mitophagy

- **Rationale.** In *Drosophila*, aneuploidy saturates autophagy, impairs mitophagy and leads to accumulation of abnormal mitochondria and senescence; enhancing protein quality control and mitophagy reduced these effects (Joy et al., 2021). Sirolimus (rapamycin) is the best-known approved inducer of autophagy and is used in children.
- **Risks.** mTOR inhibition reduces growth signalling and may worsen short stature, and it is immunosuppressive. Its relative, temsirolimus, did **not** improve outcomes when added to chemotherapy in intermediate-risk rhabdomyosarcoma (ARST1431), so no anti-cancer benefit is claimed.

Candidate 6: N-acetylcysteine — reduce oxidative stress

- **Rationale.** In the same fly model, abnormal mitochondria raise oxidative stress, which activates JNK and drives senescence; regulating oxidative stress was sufficient to reduce senescent cells (Joy et al., 2021). Candidate 5 removes damaged mitochondria; candidate 6 reduces the damage they cause.
- **Drug. N-acetylcysteine** (approved, used in children).
- **Risks.** Antioxidants accelerated progression of existing tumours in mouse models (Sayin et al., 2014; Le Gal et al., 2015). Same caution as candidate 1.

Candidate 4 (exploratory): metformin / hydroxychloroquine — stress aneuploid cells

- **Rationale.** Tang et al. (2011) found that AICAR (an AMPK activator), 17-AAG and chloroquine preferentially inhibit aneuploid cells by amplifying aneuploidy-associated stress. Approved relatives: **metformin** (AMPK activation) and **hydroxychloroquine** (autophagy/lysosome inhibitor). The idea is "cancer interception": removing aneuploid cells before they become tumours.
- **Why exploratory.** Evidence is from cell lines and uses AICAR rather than metformin. In MVA, many *normal* tissue cells are aneuploid, so selectively stressing them could worsen growth.

Hydroxychloroquine also blocks autophagy, the opposite of candidate 5. This is a trade-off to test, not a recommendation.

Table 2. Evidence grading

#	Candidate	Link to mechanism	Best evidence	Paediatric use	Tumour-promotion concern	Overall
1	NAD+ precursors	Direct (BUBR1 stability)	Mouse, BubR1-specific	Yes (vitamin; NR trial in A-T)	Moderate	**High**
2	4-PBA / ursodiol	Direct (variant-specific folding)	Mechanistic inference	Yes	Low	Medium–high
7	1 + 2 combination	Direct, genotype-derived	Inference	Yes	Moderate	Medium–high
3	Dasatinib (senolytic)	Downstream (senescence)	Mouse, BubR1-specific (genetic)	Yes (oncology)	Low–moderate; toxicity	Medium
5	Low-dose sirolimus	Downstream (proteostasis)	Fly	Yes	Low; growth concern	Medium
6	N-acetylcysteine	Downstream (oxidative stress)	Fly	Yes	Moderate	Medium
4	Metformin / HCQ	Downstream (aneuploid-cell stress)	Cell lines (proxy compound)	Yes	Low; harm to normal tissue	Exploratory

7. Considered and rejected; drugs to avoid or monitor

Rejected candidates

- **Nonsense readthrough agents** (gentamicin, ataluren) for p.Leu737Ter: the transcript is expected to undergo NMD, so little mRNA is available; ataluren's conditional EU authorisation was not renewed; aminoglycosides are nephrotoxic, a concern given nephrocalcinosis.
- **Proteasome inhibitors** (e.g. bortezomib): would block BUBR1 degradation directly but are too toxic for a child without cancer-directed indication.
- **Direct BUBR1 ligands / kinase inhibitors** (Open Targets "high-quality ligand"): would further inhibit an already deficient protein.

Avoid or use with caution

- **Growth hormone / IGF-1-axis stimulation** for short stature: increases proliferative signalling in a child with cancer predisposition.
- **Antimitotic chemotherapy (e.g. vincristine, taxanes):** these drugs rely on a functional SAC to arrest dividing cells. In a SAC-weakened background, response and toxicity may differ. We do not claim a specific effect, but recommend that the treating oncologists are aware of the BUBR1 genotype.

8. Proposed validation experiments

1. **Patient-derived cells** (fibroblasts or lymphoblastoid cells): measure BUBR1 protein level, SAC strength (mitotic arrest after nocodazole), aneuploidy rate (FISH or single-cell sequencing),

senescence markers (SA-β-gal, p16) and ROS.

2. **Test candidates 1, 2 and 7** in these cells: does BUBR1 protein increase, and does aneuploidy decrease? Check BUBR1 acetylation to test the 4-PBA interaction.
3. **Confirm the N1002K mechanism:** express wild-type and N1002K BUBR1 and compare protein half-life (cycloheximide chase) with and without chaperone.
4. **Tier 2 candidates:** in the same cells, test whether sirolimus, NAC or dasatinib(+quercetin) reduce senescence markers without increasing aneuploidy.
5. **In vivo:** BubR1 hypomorphic mice are an established model and could be used to test the lead candidates on growth, muscle mass and tumour incidence.
6. **Clinical genetics:** phase the variants by parental testing.

9. Potential impact

- **For this child:** a ranked, safety-annotated list of hypotheses that clinicians and researchers can test, plus practical cautions (growth hormone, antimitotic chemotherapy) relevant now.
- **For MVA1:** many BUB1B patients carry a missense allele in combination with a null allele. If missense alleles act by destabilising BUBR1, the “restore BUBR1 levels” strategy (candidates 1, 2, 7) may apply to a large share of MVA1 patients, and BUBR1 protein level could serve as a simple functional biomarker.
- **For other chromosomal-instability disorders:** tier 2 candidates target consequences of aneuploidy rather than the specific gene, so they may be relevant to other MVA types (CEP57, TRIP13) and related syndromes.

10. Scalability

The approach is a reusable decision framework rather than a single-case answer:

1. **Variant class → strategy.** Null alleles point to dosage restoration from the other allele or readthrough; folded-domain missense alleles point to stabilisation (chaperones, degradation control); all cases allow downstream buffering.
2. **Mechanism chain → druggable points.** The same step-by-step chain (Section 3.3) can be written for any gene.
3. **Standard evidence grading** (Table 2) with explicit tumour-promotion and paediatric-use flags, which matter for any cancer-predisposition syndrome.
4. **Public resources only** (Ensembl VEP, gnomAD, AlphaFold, Open Targets, PubMed), so the workflow can be repeated for other patients at no data cost.

11. Limitations

- No candidate has been tested in this patient’s cells; evidence comes from mice, flies, cell lines or mechanistic inference.
- The destabilising effect of N1002K is a hypothesis based on structure and related variants, not measured.
- Variant phase is not confirmed.
- Candidate identification was manual literature curation. An automated knowledge-graph check (TxGNN) could not be completed because the tool was unavailable.
- Several candidates carry a theoretical risk of promoting existing tumours; benefit–risk would need oncology input.
- No doses are proposed; paediatric dosing and drug interactions are outside the scope of this report.

12. Supporting materials

- **GitHub repository:** https://github.com/nilsugungor6450/mva-hackathon-track-2 (private during the hackathon, public afterwards). Track 2 files are in `track2/`: the evidence script, the candidate table and a README with run instructions.
- **Methods description:** the organisers' methods template, completed for Tracks 1 and 2, is in the repository (`docs/nilsu6450_methods_description_form.xlsx`).
- **Pitch video:** 3-minute walkthrough of the reasoning and candidates; link provided in the submission form.

13. Use of AI assistance

Anthropic claude.ai, Claude Opus 5.5, Claude Pro plan (consumer terms), "Help improve Claude" model-training setting turned off. Used for literature search support, explanation of biological concepts, structuring the reasoning and drafting this report. Screenshots from public databases and short excerpts of the phenotype document were shared in the chat; no genomic data files were uploaded. All candidates and claims were reviewed by me.

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